



The hidden costs of being public: Evidence from multinational firms operating in an emerging market[☆]

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ABSTRACT

I study how firms deal with business regulations that limit their operations. I first show that the ownership structure of a firm affects its degree of compliance with regulations, with publicly listed firms complying more than privately held ones. This differential compliance imposes a burden on listed firms that helps explain mergers and acquisitions patterns. When regulatory levels increase, private firms acquire listed ones and listed firms stop acquiring private ones. These results uncover an additional cost faced by listed companies, identify a new driver of M&A transactions, and show that high levels of regulation lead to opaque corporate structures.

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1. Introduction

Firms are becoming reluctant to go or remain public (Kamar et al., 2008; Gao et al., 2013; Doidge et al., 2017,

among others). Much of the existing research argues this decline responds to an increase in the costs of being listed following the enactment of regulations that only apply to public firms (e.g., the Regulation Fair Disclosure in 2000, the Sarbanes-Oxley Act in 2002, and the Global Analyst Research Settlement in 2003). In contrast, I argue that even laws and regulations that apply to both private and listed firms—such as capital control measures—impose a higher burden on listed firms and that this differential burden has increased over the last two decades following the passage and stricter enforcement of antibribery laws.¹ Listed firms are overseen by a larger number of agencies and auditors than private firms, which might translate into a higher

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¹ While the FCPA was enacted in 1977, it was not strictly enforced until the early 2000s, with two enforcement actions per year over the 1977–1996 period and 23 per year over the 1997–2017 period. Meanwhile, in countries such as Germany, bribes were not only legal but also tax deductible until 1997. These changes suggest that the indirect cost of bribing increased over time, and their timing coincides with the breakpoints in IPO patterns found in previous papers.

probability of being caught bribing officers to bypass regulations.

To support the argument that universal regulations increase the costs of compliance disproportionately, I investigate whether listed multinational firms face a differential cost of doing business in markets that impose business-unfriendly regulations, vis-à-vis similar, but private, multinational firms. I show that the listing status of a firm affects its decision to comply with regulations in an emerging market, with listed firms complying more than privately held ones. I then show that the differential cost imposed by higher compliance is so impactful that it shapes the pattern of mergers and acquisitions (M&A) transactions. In particular, following the passage of a business-unfriendly regulation, private firms acquire operations from listed ones at an extraordinary pace. My analysis uncovers an additional and significant cost faced by listed firms that operate in emerging markets, which helps explain the increasing reluctance in going public.

Testing the drivers of compliance is difficult. When firms bypass regulations, they do their best to hide their actions. I overcome this problem by focusing on a specific event that facilitates a natural mechanism by which firms may circumvent the regulation: a ban on profits repatriation. In 2012, the Argentine government banned companies from transferring funds abroad from their domestic operations. Despite this limitation, companies trying to repatriate capital could still overprice products they import from their headquarters or affiliates while keeping the retail market price fixed, thus repatriating the profits hidden in the transaction price. My detailed data set allows me to trace the transaction price for a good before and after the regulation.

Naturally, a higher postregulation price is not sufficient evidence of deliberate overpricing. The transaction price could have gone up in response to inflation in the country where the good is produced or to changes in the quality of the good, among other reasons. To overcome this identification problem, I focus on a specific industry: the automotive aftermarket (i.e., parts for replacement or repair of worn or damaged components). This industry has the peculiarity of having two different types of firms importing exactly the same goods produced by the same foreign manufacturers: (i) affiliates of the manufacturer and (ii) independent distributors that sell spare parts to end users or auto repair shops. My empirical strategy exploits this peculiarity and uses a difference-in-differences (DiD) setting to compare the transaction price for affiliates of the manufacturer and for third parties, before and after the regulation. The transaction price paid by the third party is used to control for price changes that respond to reasons other than capital repatriation since it is subject to market conditions.² Due to the granularity of the data set, I can trace and compare transactions that involve a good, identified at a very detailed level. Exploiting the richness of the setting and the data, I study whether listed and private firms exploit the overpricing mechanism to repatriate capital.

I find that after the regulation, private firms overprice imports by almost 10% and manage to repatriate up to 46% of the profits that would have otherwise remained locked in the Argentine subsidiary. Listed companies do not exploit this mechanism, showing that listing status affects compliance. I examine and reject four alternative explanations for the findings to rule out that other characteristics that might be correlated with listing status—size, ownership concentration, public visibility, and tax minimization strategies—are the drivers of the results.

One potential concern with my empirical strategy is that listing status could be endogenous. Anticipating their elusion of a regulation, firms could voluntarily delist to reduce reporting requirements. In this case, we would find that private firms' degree of compliance with regulations is lower than that of listed firms. However, causality would go in the opposite direction. In my analysis this situation is unlikely, since the kind of firm I study is the global multinational company (MNC) that operates in Argentina but is domiciled and listed abroad. Such firms would hardly change their listing status in response to a regulation in the Argentine market since it constitutes less than 0.8% of global GDP.

My methodology is free from a number of limitations of papers in the literature. Some papers use analysts' assessments of compliance,³ limiting the analysis to publicly listed companies and potentially suffering from subjectivity. Other papers use Internal Revenue Service (IRS) based measures of firms' tax evasion,⁴ but since audits are not random, sample selection bias could affect the results. I use an objective measure of compliance (the overpricing of imported goods) and analyze all of the import transactions made by both listed and private firms in a specific industry.

Having established that listed firms comply more than private ones, the next step is to test whether the magnitude of the differential compliance cost imposed by higher compliance rates is economically significant. The main empirical challenge here is that the cost is firm-time-regulation specific, and for that reason it is empirically unfeasible to measure.⁵ I take an alternative route and show that changes in the levels of market regulations impose compliance costs of such magnitude that they shape the patterns of M&A transactions. After the regulation, private firms acquired listed ones at an extraordinary pace, while listed firms stopped acquiring private ones. These patterns disappear once the ban is lifted, almost four years later, suggesting that regulation increases the cost of being public. To further test whether this pattern was a response to the regulation under study, I construct a measure of industry exposure to the regulation, denoting the percentage of earnings that each industry was repatriating before the

³ Papers such as Klapper and Love (2004) and Durnev and Kim (2005).

⁴ Papers such as Hanlon et al. (2005) and De Backer et al. (2015).

⁵ Prior attempts to estimate the cost of compliance with regulations focus on specific industries and countries. For instance, Franks et al. (1997) measure overall costs of compliance for firms in the financial services industry in the UK, and Karolyi and Taboada (2015) find evidence of regulatory arbitrage by studying cross-border bank acquisitions.

² As Lall (1973) explains, in transactions between unrelated firms each party tries to maximize its profits at the expense of the other party.

regulation. I find that global listed firms in industries more exposed to the regulations often sold their Argentine operations in the aftermath of the ban, while private global firms in the same industries did not, suggesting that sale patterns responded to the regulation under study and to the differential compliance costs that listed firms faced. It also suggests that compliance costs can be large enough to trigger the divestiture of subsidiaries.

This paper contributes to several strands of literature. First, it contributes to the literature on the costs of being public and the drivers of the going/staying public decision. Companies that go public incur significant one-time direct costs, such as underwriting and lawyers fees, as well as recurring costs, such as auditing, reporting, stock exchange fees, and compliance costs such as those associated with the Sarbanes-Oxley Act.⁶ My paper uncovers another cost: the differential cost of compliance with regulations that are not specific to listed firms. The evidence presented suggests that this cost is so great that it shapes M&A patterns. A related literature studies the determinants of the decline in the number of listed firms and focuses on changes in the economy such as technological and financial development (Gao et al., 2013 and Doidge et al., 2017, among others). My paper points out another potential factor explaining this pattern, the increased differential cost of doing business in emerging markets (EM).

This paper also contributes to the literature on determinants of M&A activity. The literature cites several drivers of M&A transactions, such as stock market misvaluations (Shleifer and Vishny, 2003; Rhodes-Kropf et al., 2005), CEO overconfidence (Malmendier and Tate, 2008), and risk diversification (Thijssen, 2008). Factors in cross-country M&A include stock market capitalization, exchange rate, differences in accounting standards, tax regimes, shareholder protection, and governance regimes of target and acquirer companies.⁷ Within this literature, the closest papers to mine are Mitchell and Mulherin (1996) and Harford (2005), who find that regulatory shocks play an important role in merger waves; Karolyi and Taboada (2015), who find evidence of regulatory arbitrage in cross-country bank acquisitions; and Maksimovic et al. (2013), who analyze the different behaviors and performance of private and listed US firms during merger waves. However, no paper in this literature seems to use the combination of regulatory shocks with firms' ownership structure to explain acquisitions.

In addition, this paper relates to the literature on the economics of crime that started with the seminal papers by Becker (1968) and Allingham and Sandmo (1972). In the last decade, literature on the determinants of corporate illegal behavior has reemerged; some link firms' characteristics with measures of compliance and governance in EM, while others estimate the firm value created by bribery. My study contains several key differences. First, I do not rely on information reported by the regulating authority (Hanlon et al., 2005; De Backer et al., 2015; Shive and Forster, 2020), whose limited capacity to detect viola-

tions might introduce selection bias. Second, I do not rely on subjective measures of compliance based on analysts' opinions (Klapper and Love, 2004; Durnev and Kim, 2005). Third, I do not use stock market prices or analysts' opinions as key inputs (Dyck and Zingales, 2004; Klapper and Love, 2004; Durnev and Kim, 2005; Karpoff et al., 2014; Zeume, 2017), which would restrict the analysis to listed firms. By including all firms (listed and private) within an industry, I can study the role of ownership structure in illicit behavior.

Last, this paper contributes to the literature on international capital flows. Several papers have looked at the effect that corruption, political risk, and regulations have on foreign direct investment (Hines, 1995; Wei, 2000a; Wei, 2000b; Smarzyska and Wei, 2000; Wei and Wu, 2002; Busse and Hefeker, 2007) and on MNCs' strategic decisions such as changes in leverage, dividends remittance, and profit allocation (Desai et al., 2004; Desai et al., 2006; Desai et al., 2008). However, to the best of my knowledge, there are no papers in this literature that analyze global firms' divestitures in response to regulatory shocks.

In summary, my paper makes five contributions. First, it uncovers an undocumented cost faced by listed companies that operate in EM: the differential cost of compliance with regulations that target all firms. Second, it identifies the ownership structure of firms as a key factor shaping M&A patterns. Third, it overcomes objectivity, data limitation, and sample selection problems and shows that the ownership structure of a firm impacts its decision on whether to comply or bypass regulations, with listed firms complying more than private ones. Fourth, it shows that an increase in the level of regulations attracts firms with more opaque corporate structures, shedding light on a new type of adverse selection problem. Fifth, it shows a novel type of market response intended to reduce the impact of regulations.

The remainder of the paper proceeds as follows. Section 2 tests whether listed firms comply more with regulations than private ones. Section 3 tests whether the differential compliance cost faced by listed firms is significant. Section 4 concludes.

2. Testing compliance rates across firms

The task of measuring compliance rates is difficult by nature. Firms do not report when they bypass regulations; therefore data are not readily available. One exception is the World Business Environment Survey (WBES) conducted by the World Bank to collect information from firms on the state of the nongovernmental sector across countries. This survey includes questions related to compliance and corruption, such as whether it is common for firms to make additional payments to do business or what the percentage of its activity is kept "off the books." However, this survey does not include information on the listing status of the respondents' firms. While other papers try to overcome this problem by using analysts' opinions, rankings, or audits data, they suffer from subjectivity, are limited to listed firms only, or can be affected by sample selection bias. To overcome these difficulties, I use a novel data set on international trade and take advantage of a

⁶ Ritter (1987); Lee et al. (1996); Engel et al. (2007); Kamar et al. (2008).

⁷ Rossi and Volpin (2004, 2007); Di Giovanni (2005); Huizinga and Volpin (2009); Siegel et al. (2011); and Erel et al. (2012).

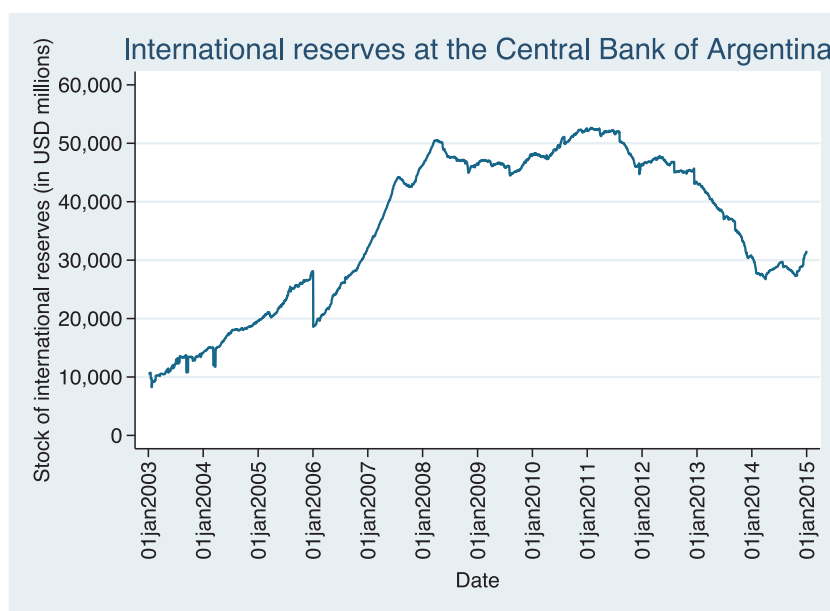


Fig. 1. Stock of international reserves at the Central Bank of Argentina. This figure reports the level of international reserves at the Central Bank of Argentina, including foreign currency, gold, and term deposits from 2003 to 2015. By the end of 2011, several macroeconomic conditions combined to cause a sharp decline in international reserves. A group of factors affected the flow of funds to and from Argentina, including the drop in international soy prices, the depreciation of the Brazilian real (which reduced the demand for Argentine goods in the Brazilian market), and the specter of inflation. Source: Central Bank of Argentina.

regulation imposed by the Argentine government in 2012 that banned companies from transferring money abroad from their Argentine operations. This setting provides an ideal natural experiment to test compliance rates since it opens a mechanism by which firms can bypass the regulation: the overpricing of imports.

2.1. The institutional setting

In 2011, several macroeconomic conditions combined to cause a sharp decline in international reserves at the Central Bank of Argentina (see Fig. 1). A group of factors affected the flow of funds to and from Argentina, including the drop in international soy prices, the depreciation of the Brazilian real (which reduced the demand for Argentine goods in the Brazilian market), and the specter of inflation. To stop the increasing outflow of capital, the Argentine government initiated a capital controls program to limit foreign currency exchange. In October 2011—a week after the elections that secured a second term for the incumbent president—the government launched the Exchange Transactions Consultation Program (Programa de Consulta de Operaciones Cambiarias) under Rule A5239. According to this measure, individuals who wanted to exchange currency had to complete a request, and the government, through the fiscal authority AFIP (its Spanish acronym), would then approve or reject the request. The decision was to be based on income, credit card balances, rent expenses, and other personal factors. This market restriction spawned the creation of a black market for US dollars, in which the foreign currency was traded at a significant pre-

mium that averaged 25% in 2012 and 60% in 2013 and peaked at 100% in May 2013, as illustrated in Fig. 2.

In November 2011, the measures were expanded to include companies' transfers. If the amount to be transferred exceeded USD 500,000, the company had to ask for authorization from the central bank. However, no request has ever been authorized by the central bank. In practice, companies were transferring amounts slightly under the threshold each day (e.g., USD 490,000) until they reached the total amount desired. In February 2012, these measures were tightened with Rule A5264. Companies were now banned from exchanging currency to transfer royalties and earnings abroad. Furthermore, a transfer of any amount required the central bank's authorization and was seldom granted. Fig. 3 plots the amount of royalties and earnings that were transferred abroad between 2008 and 2015 and shows that immediately after Rule A5264, the transfers declined by more than 94%.

The aforementioned regulation opens a natural mechanism whereby firms may circumvent the regulation. Several studies argue that firms use international trade as a mechanism to export capital. As explained by Lall (1973), when the remittance of royalties and profits are controlled, the overpricing (or transfer mispricing) mechanism provides an alternative channel for repatriation of capital. Through the purchase of overpriced goods, the buyer transfers funds to the seller. The mechanism exploited by firms to export capital can be better understood with an example (I ignore tax effects for simplicity). Suppose that firm XYZ is an MNC headquartered in the US with subsidiaries in Argentina and Mexico. Before the introduction of the regulation, the Argentine subsidiary could import

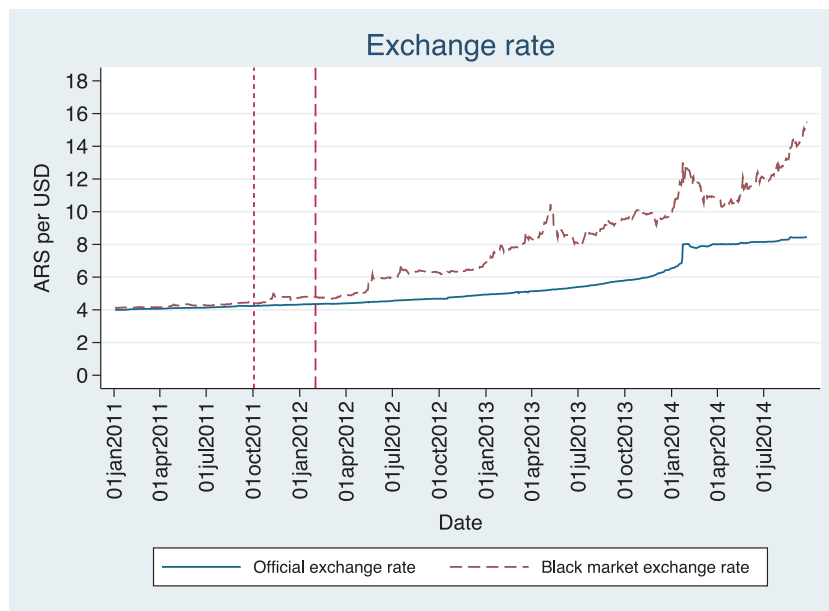


Fig. 2. Official and black market exchange rates: ARS per USD. This figure shows the evolution of both the official and the black market exchange rates for the Argentine peso (ARS) with respect to the US dollar from 2011 to 2014. The vertical lines indicate the implementation of regulations on currency exchange by individuals and profits repatriation by firms. After the Argentine government imposed restrictions on the exchange of foreign currency, the spread between the official and the black market rates increased sharply. Source: La Nación Newspaper (<http://www.lanacion.com.ar/1798393-el-dolar-blue-y-su-evolucion-en-los-ultimos-anos>).



Fig. 3. Transfer of earnings and royalties. This figure graphs the amount of US dollars that companies were repatriating from Argentina to their headquarters from 2008 to 2015. In 2011, before the ban on international transfers, companies were transferring approximately 4.5 billion US dollars per year. After the ban, the amount transferred decreased by more than 94%. The vertical line demarcates the pre and post periods. Source: Central Bank of Argentina.

goods from the Mexican subsidiary for \$10, sell them in the local market for \$15, and transfer the profit of \$5 to the headquarters, as shown in Panel A of Fig. A1 of the Online Appendix.

Facing the ban on international transfers, companies might partake in transfer mispricing, a practice that takes the form illustrated in Panel B of Fig. A1 of the Online

Appendix. In this example the Argentine subsidiary pays a higher price for the imported good to the Mexican subsidiary and makes no profit in Argentina when selling the product at the same price it was being sold before. The profit is then transferred from the Mexican subsidiary to the headquarters, thereby bypassing the effects of the regulation. However, higher postregulation transfer prices are

not sufficient evidence to claim that the company is overpricing goods to export capital.⁸ Prices might have gone up for reasons other than capital repatriation, such as inflation in Mexico or changes in the quality of the product.

To rule out alternative explanations, I use a DiD setting that compares the price paid for a good by a branch or affiliate of the manufacturer with the price paid for exactly the same good by a company unrelated to the exporter, before and after the regulation. I use the price paid by the nonrelated company as a proxy for the market price to absorb changes in price related to any macroeconomic factors common to all firms in this same industry such as inflation in the exporter country and changes in price related to product-specific factors. The granularity of the data allow me to trace each exact good imported over time and by different parties.

The decision to comply was nontrivial. Compliers would have to keep the profits made in Argentina in domestic currency and wait for the ban to be lifted to repatriate the money. Importantly, the regulation was sanctioned shortly after the president was reelected for a four-year term. An ex-post analysis shows that compliance turned out to be extremely costly. By the time the regulation was reversed (almost four years after it was imposed), the Argentine currency had depreciated against the US dollar by approximately 68%. Considering the interest rate differential on deposits between Argentina and the US, firms faced a value loss for locked-in profits of about 34%. Firms that defied regulations and exported profits by means of transfer mispricing reduced this cost but had a higher probability of being caught by the regulator and sanctioned or fined. A salient example is that of Procter & Gamble. In November 2014 the Argentine fiscal authority sanctioned Procter & Gamble and temporarily banned the company from operating in the country after accusing the company of inflating prices to take money out of the country. On top of a fine the firm had to pay, the company incurred additional costs for the lost profits during the period in which its operations were banned.

While the study of the transfer mispricing mechanism is not new, my analysis differs from the existing literature in several ways. First and most important, my paper is the first to cite the ownership structure of firms as a key determinant affecting the use of this mechanism. Second, I compare the price paid by related and nonrelated parties for the exact same good, unlike Lall (1973) who relies on experts' assessment of the value of the imported goods to proxy for the market price in Colombia. Third, unlike Bernard et al. (2006) who study intra- and extra-firm import prices for firms operating in the US and compare prices paid for products belonging to the same category, thanks to the detail in my data I can compare the price paid for the exact same good.⁹

2.2. Data and methodology

In this section, I describe the data sets and the methodology employed to study transfer mispricing by listed and privately held firms.

2.2.1. Data

I use two data sets for this section. The first data set is a novel Argentine customs database. Compared to data sets used in previous studies (e.g., the US merchandise trade database), this database offers key advantages. In addition to the information frequently available in customs databases used by researchers, the one used in the present study provides information identifying the importer (name and taxpayer ID) and product detail (brand, model, and country of manufacture) beyond what is provided by the Harmonized System code. For instance, a unique good might be an Iveco engine, model 504170202, with a capacity of 5880 cm³ and 154 kW of power manufactured by Iveco in Brazil. This level of detail allows me to compare the price paid for exactly the same product being imported by different firms at different points in time. The data set includes imports that occurred between January 2010 and September 2014, spanning two years before the regulation and two-and-a-half years after it.¹⁰

I focus on transactions belonging to the Harmonized System codes 84.07, 84.08, and 84.09,¹¹ for a key reason. Products in these categories are engine components for the automotive aftermarket (i.e., involving solely replacement or repair). Products in this industry present a peculiarity. They are imported by two different types of firms: affiliates of the manufacturer and independent distributors that sell spare parts to end users or auto repair shops. This unique characteristic of this segment allows me to compare the price paid for exactly the same product being imported by two different firms, one related and one nonrelated to the manufacturer. Therefore, the price paid by the nonrelated party can be used as a proper counterfactual since it incorporates price changes that occur for reasons other than profit repatriation.

I perform a set of restrictions to guarantee the quality of my novel data. I keep observations that include complete information on the importer, brand, and model of

even when using the maximum level of detail the system provides, the comparison of products is not precise. For instance, the ten-digit category 840733.10.30 includes "Spark-ignition reciprocating or rotary internal combustion piston engines, of a cylinder capacity exceeding 250 cc but not exceeding 1000 cc, to be installed in tractors suitable for agricultural use, not exceeding 37.3 kW." However, products within this category are hardly comparable. An engine with a capacity of 500 cc and 5kW will not have the same price as one with a capacity of 1000 cc and 37kW, even though they are grouped under the same ten-digit Harmonized System code.

¹⁰ The records of all shipments by sea, land, and air are collected by the Argentine customs authority and are made available by several vendors, such as Export Genius, Nosis, Tesys, and TradeDigits.

¹¹ These codes reference spark-ignition reciprocating or rotary internal combustion piston engines, compression-ignition internal combustion piston engines (diesel or semidiesel engines), and parts suitable for use solely or principally with the engines of heading 84.07 or 84.08, respectively. Products in these categories account for approximately 2.5% of total Argentine imports.

⁸ While this mechanism can also be used for other reasons, such as tax minimization or expropriation of minority shareholders (Djankov et al., 2008), a price change around the time of the regulation suggests that overpricing responds to the need to export capital.

⁹ Several papers identify products by their ten-digit Harmonized System code, a coding system that classifies traded products. However,

Table 1
Summary statistics.

(a) Firms							
Variable	Private			Listed			
	N	Mean	Median	N	Mean	Median	
Employees	9	56,164	8,000	20	87,061	56,089	
Revenue (million USD)	11	9,834	1,000	20	54,568	23,894	
Assets (million USD)	2	52,595	52,595	20	85,593	23,904	
Net income (million USD)	3	795	74	20	3,983	1,222	
Foreign subsidiaries	2	32	32	20	24.25	18.50	
Number of analysts	-			5	23.60	23.00	
(b) Transactions							
Variable	N	mean	p10	p25	p50	p75	p90
Unit price (FOB)	846,688	411.75	1.06	2.99	13.59	72.93	1,121.61
TX FOB price	846,688	4011.85	7.17	29.17	152.13	1,200.03	7,401.64
(c) Listing stock exchange							
		Origin	Freq.				
		NYSE	246,489				
		Toronto	42,546				
		Euronext Paris	30,411				
		Börse Berlin	25,358				
		Stockholm	23,946				
		Borsa Italiana	19,737				
		Tokio	19,350				
		Xetra	4,104				
		South Korea	3,899				
		Total	415,840				

This table presents summary statistics for the firms and products analyzed in the first section. Panel (a) presents summary statistics for the multinational companies analyzed. Available data for private multinational companies are limited; therefore the information is incomplete. Data on revenue, assets, and net income are in million USD. Panel (b) provides information on FOB prices of imported goods and transaction values. Panel (c) lists the frequency for listed firms on each stock exchange.

the product. This is required to match the importer with an exporter, determine if the two firms are related, and trace each specific product imported by different parties over time. Then, I remove observations where the importer is not a frequent importer, defined as an importer with less than 1000 units imported in the period 2010–2014, as prices for nonfrequent importers might be noisy and affect the results. I also drop transactions denominated in Argentine pesos (less than 2% of the observations), which might have been set based on an exchange rate other than the official exchange rate and are not comparable to prices set in USD. The final sample includes almost 850,000 import transactions and 73 importers, 37 of which are subsidiaries of MNCs, and 36 of which are independent domestic companies. Among the subsidiaries, 23 are subsidiaries of listed companies and 14 are subsidiaries of private companies; all of the parent companies are incorporated in countries that belong to the Organisation for Economic Co-operation and Development (OECD). Since some importers share the same ultimate parent company, the final sample has 20 listed MNCs and 13 private MNCs. Altogether, these firms import products from 88 different countries.

The summary statistics provided in Table 1 describe the data. The first panel provides information on the MNCs in the sample when available (since private firms do not necessarily report their data, information is incomplete for this subset of firms). While private and listed MNCs

appear to be quantitatively different in several observable dimensions, the available information shows that both private and listed MNCs are very large. Private MNCs have, on average, 56,164 employees and revenues of almost USD 10 billion, while the numbers for listed MNCs are 87,061 and USD 54 billion, respectively. Panel B summarizes import data at the transaction level, where each transaction indicates a line in a bill of lading. The mean (median) unit value of imported goods is USD 411.75 (USD 13.59), and the mean (median) value of each line is USD 4,011.85 (USD 152.13). Panel C details the listing stock exchange for imports made by listed firms.

The second data set I use is the World Intellectual Property Organization (WIPO) Global Brand Database (available at www.wipo.int) that includes information on registered trademarks. Since the imports data set does not include information on the identity of the exporter, I match each imported product with its manufacturer by looking at the firm that owns the trademark in the country from which the product is being exported. For instance, if I observe a product “Veglia Borletti” being imported from Indonesia, I search for the trademark and find that the owner of that brand in that country is the company Magneti Marelli S.p.A. It could be argued that the firm producing the good might not be the firm exporting it. However, for the present study, it is sufficient to know that there is a relationship between the importer and the producer. The

presence of an intermediary does not affect the potential use of the transfer mispricing mechanism.

The next step is to determine whether the firm importing the goods in Argentina and the exporter for each transaction are related. To that end, I manually search for information on each firm on the internet and determine whether it is a multinational company and who the parent company is. As an example, by looking at Magneti Marelli S.p.A., I find that the company's parent organization is Fiat S.p.A. Next, I look at each transaction, determine whether the importer and the exporter are branches or affiliates under the control of the same parent firm, and create an indicator for related firms. For instance, if the importer of the Veglia Borletti product is Fiat Argentina, I flag the transaction as between related parties since Fiat Argentina (importer) and Magneti Marelli (exporter) belong to the same parent company. Last, I manually search for information on the ownership structure of the parent company, with respect to whether it is listed or private. In this case, I find that at the time of the transaction, Fiat S.p.A. (or Fiat Industrial S.p.A.) was listed on the Milan stock exchange.

2.2.2. Methodology

I exploit a regulation imposed by the Argentine government in 2012 banning companies from transferring funds from Argentine operations abroad as a source of exogenous variation, and I use firms' ownership structure as a source of cross-sectional variation. Companies trying to repatriate capital despite the regulation could overprice products imported from their headquarters or affiliates. However, higher postregulation prices could be caused by reasons other than capital repatriation, such as inflation or changes in the quality of the product. Therefore, a simple comparison of prices before and after the regulation is not sufficient.

I focus on a specific industry: the automotive aftermarket, or parts for replacement or repair of worn or damaged components. This industry has a characteristic that is key to overcoming the problem mentioned before. In this industry, two types of firms import exactly the same goods produced by the same foreign manufacturer in the same country: (i) affiliates of the manufacturer and (ii) independent distributors that sell spare parts to end users or auto repair shops. I exploit this peculiarity and use a DiD setting to compare the price paid for a product by a related importer with the price paid by a nonrelated importer, before and after the regulation.

A DiD setting requires a subset of the units under study to be subject to a treatment, while the other subset is used as a control. Since in my setting every firm is subject to the same treatment (i.e., the ban on international transfers), the usual definition of treated and control firms cannot be applied. However, the context allows for an interesting definition of treated and control firms. In this setting, a treated unit is one that imports products from its headquarters or an affiliate manufacturer, defined as a manufacturer that belongs to the same MNC. Meanwhile, a control unit is one that imports products from a nonrelated party. Comparing the price paid by a related party with the one paid by a nonrelated party allows me to isolate the effect that factors other than capital

repatriation motives might have on the price of a product. Since the price paid by a nonrelated party is subject to market conditions, it includes all of those potential factors. The extraordinary granularity of the data set allows me to trace a product identified with extreme detail and to compare the price paid for exactly the same good imported by two different parties at different points in time.

In addition to the two standard dimensions used in a DiD setting—treated and post—I exploit a third source of variation: treated firms' listing status. The literature on the economics of crime has shown that a key driver of the decision to evade a norm is the probability of being detected. Since listed firms are subject to stricter scrutiny than private firms—for instance, foreign subsidiaries of US-listed firms are audited by a third-party—the conditional probability of their being detected is greater. Thus, I exploit the listing status of the MNC's parent company as the cross-sectional source of variation for testing whether private and listed firms behave differently. To this end, I divide the treated group—which imports goods from related parties—into two different subgroups: (1) subsidiaries of MNCs whose parent companies are listed and (2) subsidiaries of MNCs whose parent companies are private. Importantly, since the firms used as controls are independent distributors, in all of the cases they are private Argentine firms. To facilitate interpretation, I present the results in a DiD setting, splitting the sample between listed and private firms. However, in the Online Appendix I provide the results in a triple differences setting. The results are similar to the ones presented here.

Analyzing MNCs serves two purposes. First, MNCs have a natural advantage at exploiting the transfer mispricing mechanism. Firms with no affiliates need a third party to act as an accomplice and to pay the difference between the market price and the paid price in a foreign account, without leaving a record. Second, analyzing MNCs imparts exogeneity to the listing status of firms since an MNC would hardly change its global listing status in response to a regulation in Argentina.

The main regression specification used in this section is

$$P_{ijkt} = \alpha_i + \alpha_m + \beta_p \times Post + \beta_r \times Related_{jk} + \beta_{pr} \times Post \times Related_{jk} + \epsilon_{ijkt}, \quad (1)$$

where i represents the imported good, j indicates the importer, k indicates the exporter, m indicates month of the year, and t indicates time. The variable under study is the price paid for the good i being imported by firm j and exported by firm k at time t . To keep the study uncontaminated by changes in other costs such as insurance, freight, and shipping costs, I use the free on board price, the price of the good ready to be shipped at the origin port. The exporter pays for transportation of the goods to the port of shipment, plus loading costs. In addition, the import price is normalized by the mean price of that product to provide equal consideration to products at different price levels. The specification includes product (α_i) and month (α_m , January to December) fixed effects. Product fixed effects control for all time-invariant characteristics of the product and for all time-invariant characteristics of the exporter since each product is manufactured by only one company.

Month fixed effects control for potential seasonality. The standard errors are clustered at the importer-exporter pair level (682 pairs), and the results are robust to clustering at the exporter level.

The coefficient β_r is expected to be negative. This coefficient is related to the price wedge in Bernard et al. (2006), where the authors show that US exporters set lower prices for related importers than for arm's-length importers due to longer-term contracts, managerial incentives, or tax reasons. The coefficient β_p indicates the price change for products after the regulation. If demand for imported products is reduced, we might expect a negative coefficient to increase sales. However, if exporters expect delays in liquidation of international payments, they might increase prices to account for a longer financing period. Therefore, the sign of this coefficient cannot be predicted.

The coefficient of interest is β_{pr} , which indicates whether the price change after the regulation for a related importer is different than that for a nonrelated importer. If companies start paying a higher price for products imported from related parties to export capital, we should expect β_{pr} to be positive. The magnitude of the coefficient is a priori unknown and should respond to a trade-off. On the one hand, higher price manipulation allows a company to export a higher fraction of the profits that cannot be exported anymore via an international transfer. On the other hand, the higher the manipulation, the higher the probability of being detected by the regulator.

The DiD design requires that treated and control firms would follow parallel trends had the regulatory intervention not happened. To assess the plausibility of this untestable assumption, I test whether the two groups of firms followed parallel trends before the regulation change. Fig. 4 plots the deviation from the demeaned price for goods imported by both firms related and non-related to the exporter. In addition, I include the series for private-related importers only. Before the regulation, the prices for related and nonrelated importers move together and the price for a related importer is lower than the price for a nonrelated importer, which is consistent with prior findings in the literature. At the time of the regulation, the patterns for related importers seem to change, but they do not exhibit a clear pattern consistent with imports overpricing. However, when I exploit the third source of variation—listing status of the MNC—and analyze the series for private related importers only, the pattern is striking. At the time of the regulation, prices for private-related importers increase significantly, while prices for nonrelated importers decrease. This evidence suggests that private firms overprice imports to export capital.

An alternative explanation for the pricing patterns would be significant changes in the volumes imported since they might affect bargaining power or discounts given by exporters. Fig. 5 plots the aggregate volume of imports by domestic, listed, and private MNCs for these products. Panel (A) shows total imports from all types of exporters, while panel (B) shows only imports from related exporters. Both panels show stable patterns, suggesting that volumes cannot explain the pricing patterns observed.

As for control variables, we might control for time-varying characteristics of the firms. Unfortunately, information on the Argentine operations of firms is protected by national fiscal secrecy rules. As for global private MNCs, they do not disclose financial statements. In contrast, global listed MNCs disclose their financial statements; however the statements are consolidated, and do not allow for the isolation and study of Argentine subsidiaries' operations. For this reason, I conduct several robustness tests to rule out alternative explanations such as differences in size, concentration of ownership, or visibility.

2.3. Results

I first show that private MNCs have a lower compliance rate than listed MNCs, participate in transfer mispricing, and export a significant fraction of the profits via this mechanism. Then I present several robustness tests to rule out alternative explanations.

The results of regression specification 1 are presented in Table 2, where the sample is split between transactions made by private MNCs (columns [1] and [2]) and listed MNCs (columns [3] and [4]). In both cases, approximately 360,000 transactions made by domestic firms are included as controls. The coefficient on *Related* shows that exporters set a lower price for related importers, which is consistent with findings in Bernard et al. (2006). The difference calculated in column [5] shows that this practice does not differ between listed and private firms, ruling out alternative explanations such as the differential use of tax avoidance strategies by listed or private firms.

More importantly, the results provided in the first column of Table 2 show that after the regulation, the price increase for a product imported by a related company is 9.96% higher than that for a nonrelated company if the related importer is a private company. These results are significant at the 1% level. In column [2] the specification includes month fixed effects to control for potential seasonality issues, and the results are not altered. The evidence suggests that private companies are manipulating transfer prices to export capital. Results in columns [3] and [4] measure price increases for listed firms relative to that of nonrelated importers. The differential price increase is indistinguishable from zero, suggesting that listed firms do not exploit the transfer mispricing mechanism. The difference in the price increase between private and listed MNCs relative to the difference in price increase for nonrelated importers is shown in column [5]. The difference between the two coefficients for *Post × Related* is significant at the 1% level. In contrast, the differences in the individual coefficients for *Post* and *Related* are not significant at the 10% level.

The quarter-by-quarter coefficients plotted in Fig. 6 show that private firms respond immediately. As a matter of fact, private firms seem to anticipate the regulation and increase prices a quarter before it is enacted, when transfers for individuals where banned and transfers for companies were limited. In contrast, listed firms do not change prices.

To understand the magnitude of the coefficient β_{pr} , I do a back-of-the-envelope calculation using information

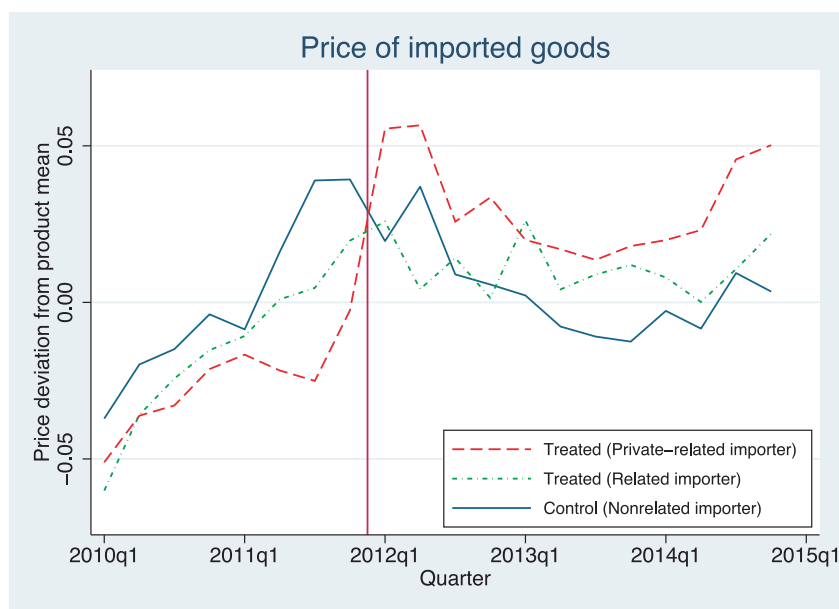


Fig. 4. Price paid by related and nonrelated importers, before and after the regulation. This figure plots the deviation from the demeaned price for products imported by both related and nonrelated importers from 2010 to 2014. In addition, I include the series for private-related importers only. Before the regulation, related importers paid lower prices than nonrelated importers. At the time of the regulation, prices for private-related importers increased significantly compared to prices for nonrelated importers, suggesting that companies were overpricing imports to export capital. The vertical line demarcates the pre and post periods.

Table 2
Transfer mispricing mechanism.

Price	Private firms		Listed firms		Difference
	(1)	(2)	(3)	(4)	
Post	0.00376 (0.020)	0.00172 (0.020)	0.00861 (0.017)	0.00810 (0.017)	−0.00485
Related	−0.0530*** (0.018)	−0.0548*** (0.017)	−0.0889** (0.039)	−0.0892** (0.039)	0.0359
Post × Related	0.0996*** (0.031)	0.0990*** (0.031)	0.0134 (0.017)	0.0133 (0.018)	0.0862***
Product FE	Yes	Yes	Yes	Yes	
Month FE (Jan-Dec)	No	Yes	No	Yes	
N	430,846	430,846	787,591	787,591	
R-squared	0.043	0.044	0.004	0.005	

This table presents the import prices in a DiD setting. The sample consists of imported goods belonging to Harmonized System codes 84.07 (Spark-ignition reciprocating or rotary internal combustion piston engines), 84.08 (Compression-ignition internal combustion piston engines (diesel or semidiesel engines)), and 84.09 (Parts suitable for use solely or principally with the engines of heading 84.07 or 84.08) from January 2010 to September 2014. The dependent variable is the import price normalized by the mean price of that product, which is done to give equal consideration to products at different price levels. *Post* is a dummy indicator for import transactions that occurred after Rule A5264. *Related_{jk}* is a dummy indicator for transactions between importers and exporters that belong to the same MNC. The results in columns [1] and [2] measure the differential price increase for private firms, using the price paid by unrelated domestic firms as a control. The results in columns [3] and [4] measure the same differential price increase for listed firms. Column [5] shows the difference in the coefficients obtained from columns [1] and [3]. Standard errors are clustered at the importer-exporter pair level and are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

obtained from one of the companies in the sample. Before the regulation, this firm had a ratio of cost of goods sold (COGS) to sales of 84% and a ratio of selling, general, and administrative expenses to sales of 9.5%. Ignoring taxes for the sake of simplicity, this firm would have a pretax earn-

ings to sales ratio of 6.5%. Lacking extensive information and proceeding conservatively, based on conversations with firms in the industry, I assume the FOB price of the imported goods to be 30% of sales, with the balance of 54% corresponding to assembling, shipping, packaging, and

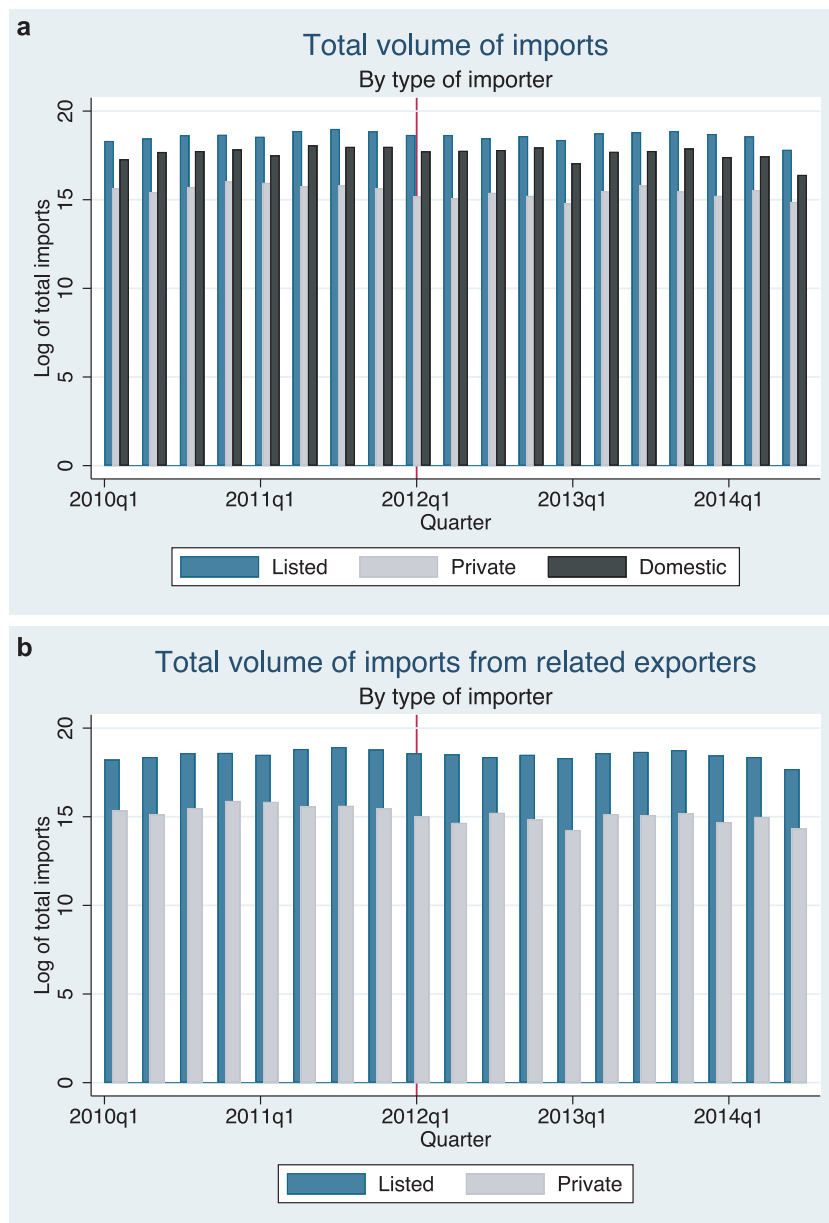


Fig. 5. Volume of imports. This figure plots the total volume of imports (taken from a log of FOB prices) made by firms in my sample from 2010 to 2014. Panel (A) includes products imported from all types of exporters. Panel (B) includes only products imported from related exporters. The vertical line demarcates the pre and post periods. This evidence suggests that the volumes imported did not change significantly after the regulation and were similar for both listed and private firms.

other costs. For a company with these characteristics, the 10% overprice of imports would represent a transfer of 46% of the pretax profits earned since 10% of a cost that amounts to 30% of sales represents 3% of sales, or 46% (3% divided by 6.5%) of the pretax profits. This magnitude can be thought of as a lower bound since the amount of money transferred via overpricing of imports would represent a larger fraction of posttax profits.¹² The following table summarizes the back-of-the-envelope exercise.

	Before	After
Sales	100	100
COGS		
FOB	(30)	(33)
Others	(54)	(54)
SG&A	(9.5)	(9.5)
Pretax profits	6.5	3.5

2.4. Robustness checks

In the next sections, I describe the robustness tests performed to rule out alternative explanations for the results.

¹² I thank Gordon Bodnar for this comment.



Fig. 6. Quarter-by-quarter evolution of prices paid by private-related and listed-related importers. This figure plots the quarter-by-quarter regression coefficients for products imported by both private-related and listed-related importers. When the government imposed limits on profits repatriation—a quarter before banning them—private-related importers increased prices to export capital, while listed-related importers did not.

2.4.1. Size

Listed firms are, in general, larger than private ones, and larger firms might get more attention from regulators than smaller ones. Thus, it could be argued that it is size—and not listing status—that affects firms' compliance with business-unfriendly regulations. To test this theory, I construct a subsample of listed firms that are comparable in size to private firms and repeat the analysis for this subset of firms. Since unconsolidated data on the size of the Argentine operations for each firm are not available, I proxy it with the total volume of dollar imports before the regulation. The results in Table 3 show that listed firms that are comparable to private firms do not overprice imports to export capital. This result suggests that size does not affect behavior regarding the ban on profits repatriation.

2.4.2. Concentrated ownership

It might also be argued that listed firms in general have a more dispersed ownership structure and this characteristic might influence political connectedness, risk aversion, or the ability to coordinate actions. To rule out this alternative explanation, I split listed firms into those that have a shareholder with 10% or more of the shares and those that do not, taking after La Porta et al. (1999). I repeat the study for the subset of firms that have concentrated ownership to test whether this is the factor driving non-compliance. The results in Table 4 suggest that listed firms with concentrated ownership do not exploit the overpricing mechanism. These results rule out the alternative explanation that ownership concentration drives the findings.

2.4.3. Visibility

In addition to being larger and having more dispersed ownership, listed firms are usually more publicly visible than private firms. Therefore, it might be that the more visible firms comply more with regulations. To test whether visibility drives compliance, I construct a measure of visibility using the Factiva data set to determine the number of news items that mention each company in 2011—the year before the regulation to avoid potential endogeneity problems—and repeat the test, splitting listed firms into those that are more visible (visibility level above the median) and those that are less visible (visibility level below the median). The results in Table 5 show that visibility is not a factor that drives compliance. While it would be interesting to test how the characteristics analyzed interact to affect the results, the relatively small number of firms in my sample does not allow me to exploit the effect of the interaction of firm-level characteristics, such as small firms with concentrated ownership or visible firms with disperse ownership.

2.4.4. Corporate taxes and tariffs minimization strategy

The DiD setting is ideal for controlling for shocks that affect all firms. However, there are also shocks that affect some firms and not others, such as changes in exporting countries' tax rates or changes in tariffs. Consider a case in which the corporate tax rate in Brazil decreases. Since transfer prices can also be used to shift profits from one subsidiary to another, firms that import goods from their affiliates in Brazil would have incentives to increase transfer prices and to reduce the overall amount of tax the

Table 3

Transfer mispricing mechanism: size.

Price	Private firms		“Small” listed firms		Difference
	(1)	(2)	(3)	(4)	
Post	0.00376 (0.020)	0.00172 (0.020)	0.00860 (0.017)	0.00769 (0.017)	–0.00484
Related	–0.0530*** (0.018)	–0.0548*** (0.017)	–0.122** (0.061)	–0.122** (0.061)	0.069
Post × Related	0.0996*** (0.031)	0.0990*** (0.031)	0.0125 (0.018)	0.0122 (0.018)	0.0871**
Product FE	Yes	Yes	Yes	Yes	
Month FE (Jan-Dec)	No	Yes	No	Yes	
N	430,846	430,846	686,319	686,319	
R-squared	0.043	0.044	0.007	0.008	

This table presents the import prices in a DiD setting and compares the behavior of private firms with that of listed firms of comparable size from January 2010 to September 2014. The dependent variable is the import price normalized by the mean price of that product, which is done to give equal consideration to products at different price levels. *Post* is a dummy indicator for import transactions that occurred after Rule A5264. *Related_{ijk}* is a dummy indicator for transactions between importers and exporters that belong to the same MNC. The results in columns [1] and [2] measure the differential price increase for private firms, using the price paid by unrelated domestic firms as a control. The results in columns [3] and [4] measure the same differential price increase for listed firms comparable in size to private firms. Column [5] shows the difference in the coefficients obtained from columns [1] and [3]. Standard errors are clustered at the importer-exporter pair level and are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 4

Transfer mispricing mechanism: concentrated ownership.

Price	Private firms		Listed firms w/ concentrated ownership		Difference
	(1)	(2)	(3)	(4)	
Post	0.00376 (0.020)	0.00172 (0.020)	–0.00538 (0.019)	–0.00634 (0.019)	0.00914
Related	–0.0530*** (0.018)	–0.0548*** (0.017)	–0.109** (0.052)	–0.109** (0.052)	0.056
Post × Related	0.0996*** (0.031)	0.0990*** (0.031)	0.0292 (0.020)	0.0292 (0.021)	0.0704***
Product FE	Yes	Yes	Yes	Yes	
Month FE (Jan-Dec)	No	Yes	No	Yes	
N	430,846	430,846	693,535	693,535	
R-squared	0.043	0.044	0.010	0.011	

This table presents the import prices in a DiD setting and compares the behavior of private firms with that of listed firms with concentrated ownership from January 2010 to September 2014. The dependent variable is the import price normalized by the mean price of that product, which is done to give equal consideration to products at different price levels. *Post* is a dummy indicator for import transactions that occurred after Rule A5264. *Related_{ijk}* is a dummy indicator for transactions between importers and exporters that belong to the same MNC. The results in columns [1] and [2] measure the differential price increase for private firms, using the price paid by unrelated domestic firms as a control. The results in columns [3] and [4] measure the same differential price increase for listed firms with a concentrated ownership structure, defined as having a shareholder that owns at least 10% of the shares. Column [5] shows the difference in the coefficients obtained from columns [1] and [3]. Standard errors are clustered at the importer-exporter pair level and are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

company pays. If only private MNCs imported goods from Brazil, listed MNCs would not be able to exploit this mechanism even if they wanted to. If listing status is correlated with changes in tax rates or tariffs, my findings could be responding to the possibility, rather than intentionality, of firms bypassing regulations.

To rule out this alternative, I control for changes in corporate tax rates for each country that Argentine firms in my sample are importing from and for changes in tariffs. For this, I include a different dummy variable set to one for each country-year with a change in the corporate tax rate to absorb price changes that respond to tax minimization

strategies and exclude transactions corresponding to Harmonized System subcategories that were subject to tariff changes over the sample period (less than 1% of the transactions). The results of this specification are provided in Table 6 and show that the differential price increase does not correspond with tax minimization strategies nor with changes in tariffs.

2.4.5. Other avenues

Overpricing imports is one of several mechanisms that firms might use to export capital in spite of the regulation. Firms can also overpay for services at their headquarter-

Table 5

Transfer mispricing mechanism: visibility.

Price	Private firms		Listed firms w/ low visibility		Difference
	(1)	(2)	(3)	(4)	
Post	0.00376 (0.020)	0.00172 (0.020)	0.00838 (0.018)	0.00721 (0.018)	–0.00462
Related	–0.0530*** (0.018)	–0.0548*** (0.017)	–0.201* (0.105)	–0.202* (0.105)	0.148
Post × Related	0.0996*** (0.031)	0.0990*** (0.031)	0.0215 (0.023)	0.0216 (0.023)	0.0781**
Product FE	Yes	Yes	Yes	Yes	
Month FE (Jan–Dec)	No	Yes	No	Yes	
N	430,846	430,846	523,762	523,762	
R-squared	0.043	0.044	0.012	0.013	

This table presents the import prices in a DiD setting and compares the behavior of private firms with that of listed firms with low visibility from January 2010 to September 2014. The dependent variable is the import price normalized by the mean price of that product, which is done to give equal consideration to products at different price levels. *Post* is a dummy indicator for import transactions that occurred after Rule A5264. *Related_{jk}* is a dummy indicator for transactions between importers and exporters that belong to the same MNC. The results in columns [1] and [2] measure the differential price increase for private firms, using the price paid by unrelated domestic firms as a control. The results in columns [3] and [4] measure the same differential price increase for listed firms with low visibility, defined as having a below median number of mentions in Factiva. Column [5] shows the difference in the coefficients obtained from columns [1] and [3]. Standard errors are clustered at the importer-exporter pair level and are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 6

Transfer mispricing mechanism: tax and tariff minimization strategies.

Price	Private firms		Listed firms		Difference
	(1)	(2)	(3)	(4)	
Post	0.00337 (0.0202)	0.00129 (0.0204)	0.00831 (0.0168)	0.00793 (0.0170)	–0.00494
Related	–0.0549*** (0.0186)	–0.0571*** (0.0178)	–0.0972** (0.0412)	–0.0972** (0.0412)	0.0423
Post × Related	0.0978*** (0.0288)	0.0962*** (0.0289)	0.0198 (0.0174)	0.0190 (0.0175)	0.078***
Product FE	Yes	Yes	Yes	Yes	
Month FE (Jan–Dec)	No	Yes	No	Yes	
Corporate taxes	Yes	Yes	Yes	Yes	
N	426,378	426,378	779,998	779,998	
R-squared	0.0428	0.0445	0.0071	0.0076	

This table presents the import prices in a DiD setting and includes dummy variables that are set to one for each exporting country-year with a change in corporate tax rates and excludes transactions in product subcategories that experienced changes in tariff rates across the sample period. The sample period is from January 2010 to September 2014. The dependent variable is the import price normalized by the mean price of that product, which is done to give equal consideration to products at different price levels. *Post* is a dummy indicator for import transactions that occurred after Rule A5264. *Related_{jk}* is a dummy indicator for transactions between importers and exporters that belong to the same MNC. Columns [1] and [2] measure the differential price increase for private firms, using the price paid by unrelated domestic firms as a control. Columns [3] and [4] measure the same differential price increase for listed firms. Column [5] shows the difference in the coefficients obtained from columns [1] and [3]. Standard errors are clustered at the importer-exporter pair level and are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

ters or inflate the interest on intra-firm loans. Even though firm-level data are not available, a close look at aggregated data seem to suggest that firms do not exploit these avenues.¹³ Fig. 7 plots the transfers abroad made by firms in Argentina with regard to (i) payment for services such as professional and technical services, insurance, communica-

tion services, IT services, IP and trademark, and copyright; and (ii) interest payments.

Payment for services does not increase after the regulation, suggesting that firms did not use this avenue to export capital. In contrast, interest payments do increase after the regulation. However, this increase can be explained by an increase in the credit risk spread of Argentina, as measured by J.P. Morgan's Emerging Markets Bond Index (EMBI+) for Argentina.

In addition, listed firms could enter into private agreements with the government to be excluded from the ban.

¹³ The maximum level of disaggregation is the industry level, which does not allow the comparison of transfers made by listed and private firms.

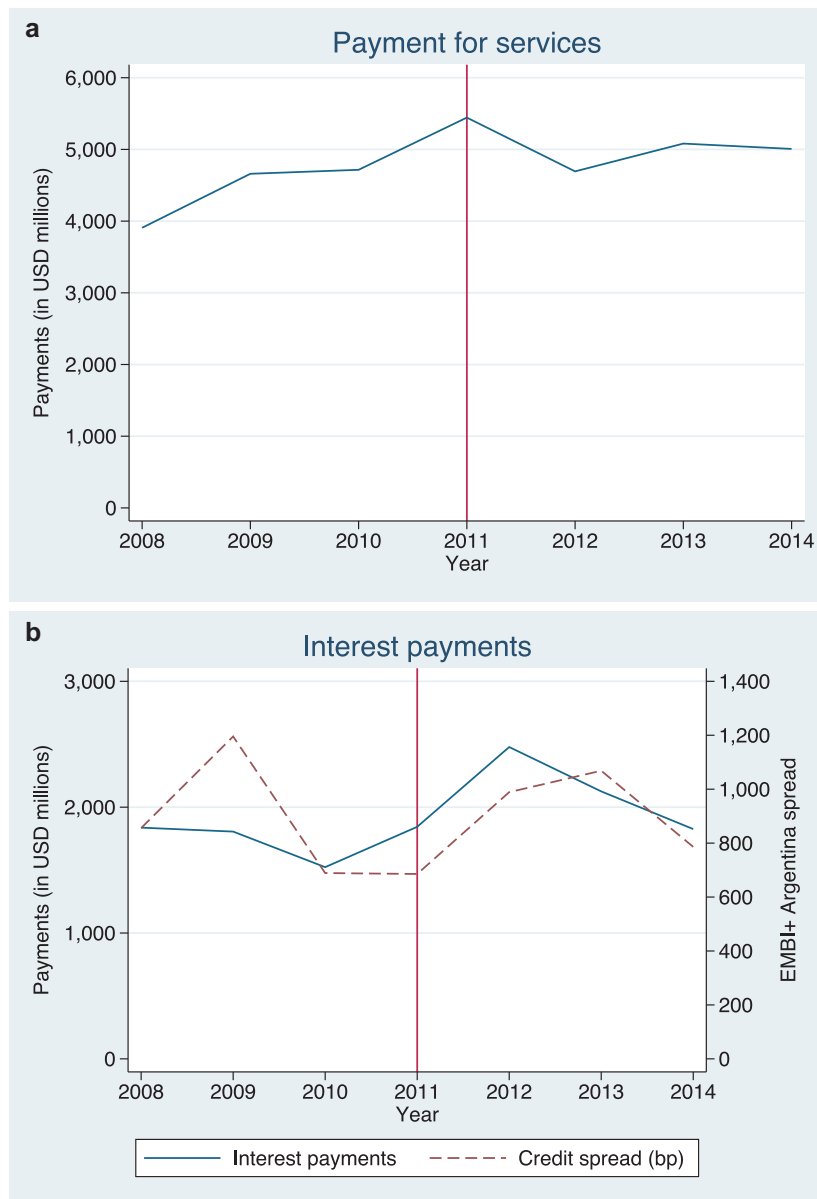


Fig. 7. Alternative channels. These graphs plot transfers abroad made by firms in Argentina, with regard to payment for services (Panel (A)) and interest payments (Panel (B)) from 2008 to 2014. The graph shows that after the regulation, payment for services did not increase, suggesting that firms did not use this avenue to repatriate capital. In contrast, interest payments did increase after the regulation. However, the increase seems to correspond with higher interest rates, as reflected by the increase in the credit spread as measured by J.P. Morgan's Emerging Markets Bond Index (EMBI+) for Argentina. The vertical line demarcates the pre and post periods. Source: Central Bank of Argentina and Ambito Financiero Newspaper.

For instance, in Luxembourg the official corporate tax rate is 29%. Yet, several firms have private agreements with the tax authority and pay rates close to 0%.¹⁴ If this were the case, listed firms would not need to overprice imports. However, Fig. 3 suggests that the ban affected all firms.

Although other avenues might be available, the fact that listed firms do not overprice imports—a mechanism that is readily available to them—and do not seem to

use other mechanisms that are easy to implement suggests that it is costlier for listed firms to bypass regulations. The tax authorities can and potentially do detect the use of this mechanism. Thus, a potential explanation for the differences found in compliance can be attributed to the differential scrutiny listed firms are subject to. For instance, while both listed and private firms could “grease the wheels” to bypass regulations in EM, they can be prosecuted in their home countries for doing this. Differences in the strictness of auditing or likelihood of prosecution between firm types could then drive the results.

¹⁴ I thank Stefan Zeume for this comment.

3. The impact on the M&A market

The previous section shows that private firms have more flexibility than listed ones in terms of using mechanisms that assist them in bypassing a regulation. In this section, I assess the magnitude of the cost differential in compliance with regulations. This assessment presents two empirical challenges. The first empirical challenge is that the traditional event study methodology cannot be applied in this setting for two reasons: i) we do not observe market value for private firms, and ii) for listed MNCs, we should not expect a significant change in their value after a regulation in an EM such as Argentina because the size of the Argentine operations are relatively small compared to global operations. The second empirical challenge is that the differential cost of compliance is firm-time-regulation specific, with several factors rendering it unobservable.

To show that the cost differential is large, I take an alternative route and show that changes in the levels of market regulations impose differential compliance costs of such magnitude that they shape the patterns of M&A transactions. The rationale is this: if in business-unfriendly markets listed firms face a much higher compliance cost than private ones, after a sudden increase in business-unfriendly regulations, we should expect private firms to acquire listed ones since the former can mitigate the regulatory burden and increase the value of a listed firm. In this section, I show that this happened in Argentina, where the aforementioned regulation took place. In addition, I show that this pattern does respond to the ban on profit repatriation and once the ban was lifted, the pattern disappears.

3.1. M&A transactions

The regulation exploited in the previous section had important consequences for firms operating in Argentina. An article in *The Economist* titled “Waiting for Cristina to go” and dated December 2014¹⁵ states that in response to harshening business conditions and according to a study by a consultancy firm, “40 foreign companies have left or frozen their operations since 2011.” Notably, the ban on profits repatriation was imposed shortly after the national elections that secured a second term for the incumbent president, Cristina Fernandez de Kirchner, who obtained 54% of the votes, while the runner up obtained less than 17% of them. Thus, while the duration of the ban was uncertain, there were no expectations that it would be lifted in the short or medium term. In the context of this paper, a foreign firm or MNC leaving the country would be reflected in an M&A transaction wherein the MNC sells its Argentine subsidiary. In line with the findings of the previous section, since private firms can significantly mitigate their regulatory costs, one might expect that listed firms sell their operations to private firms, while listed firms would stop acquiring private ones.

To test this conjecture, I first expand my list of firms by incorporating all of the transactions in the Bureau

Van Dijk’s Zephyr M&A data set where the Argentine subsidiary of an MNC—the group of firms most affected by the regulation—was sold between years 2000 and 2015. I do not focus exclusively on M&A transactions in the automotive market for two reasons. First, there is a limited number of M&A in this industry. Second, analyzing multiple industries allows me to exploit heterogeneity in exposure to the regulation across industries. I start with an initial data set of more than 2,200 M&A transactions, out of which approximately 80% involve the selling of the Argentine subsidiary of an MNC. As is customary in the literature, transactions where the target company belongs to a regulated industry such as utilities (SIC codes 4900–4949) or financial institutions (SIC codes 6000–6999) are excluded. My final data set comprises 1,492 M&A transactions across multiple industries over the period 2000–2015. In addition, since the focus of this paper is on firms that leave the country by selling their operations, for robustness I repeat the tests employing only transactions where the final stake of the acquirer is greater than or equal to 95%, resulting in a subsample of 762 transactions. This threshold accounts for the existence of minority shareholders, but the results are robust to using other thresholds.

For each year, I calculate two values: (i) the percentage of transactions in which the seller is a listed MNC and the acquirer is a private firm and (ii) the percentage of transactions in which the seller is a private MNC and the acquirer is a listed firm. In panel (A) of Fig. 8, I plot both types of transactions and show patterns consistent with the conjecture above. More specifically, after the regulation, private firms (which can bypass regulations and reduce the burden of compliance) started acquiring listed firms at an extraordinary pace, while listed firms stopped acquiring private ones altogether. Since the acquirer can be a domestic or a foreign firm, to show that the domicile of the company does not drive the results, in panel (B) I plot the percentage of transactions where the acquirer is a domestic firm across time. The pattern suggests that the domicile of the acquirer does not drive the results.¹⁶ In addition, in panel (A) of Fig. 9, I show that the total number of transactions in the period after the regulation does not differ significantly from that of the period prior to the regulation. In unreported results I also find that the percentage of transactions wherein an MNC sells its subsidiary over the total number of M&A transactions in Argentina and the percentage of transactions wherein an MNC acquires a domestic firm also do not change significantly during the sample period and do not seem to follow any pattern that might drive the results. Last, as a placebo, I search for patterns in M&A transactions between private acquirers and private targets and between listed acquirers and listed targets. Panel (B) of Fig. 9 shows the absence of similar patterns.

¹⁵ <https://www.economist.com/business/2014/12/11/waiting-for-cristina-to-go>, accessed on July 10, 2018.

¹⁶ While the regulation was potentially costlier to foreign firms than to domestic ones and a larger fraction of transactions with domestic acquirers might have been expected, in case an MNC sells its Argentine subsidiary to a domestic firm, the proceeds of the transaction would still have been locked in in Argentina, making it preferable for the seller to sell the company to another (private) MNC and having the transaction payment made outside of Argentina.

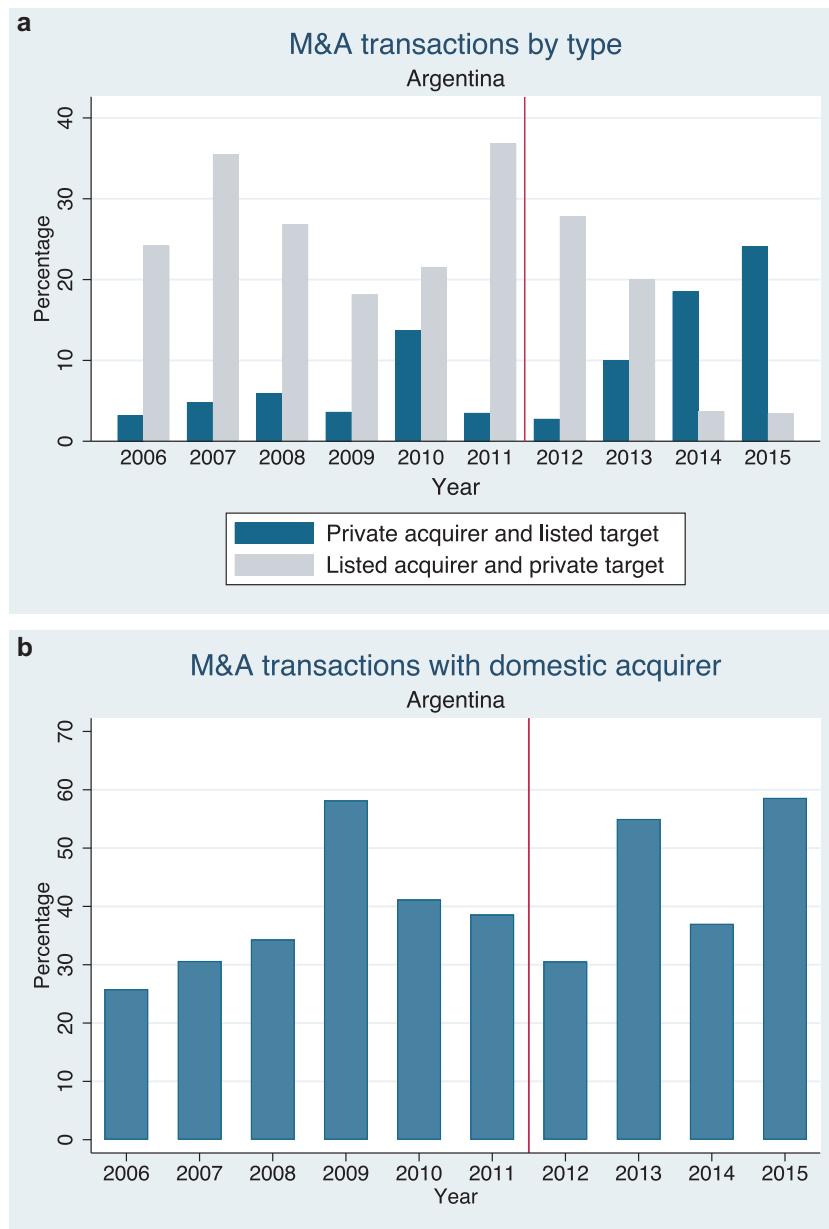


Fig. 8. Type of transactions before/after ban on international transfers. These graphs plot M&A transactions wherein an MNC sells its Argentine subsidiary, between the years 2006 and 2015. Panel (A) shows the percentage of the total number of transactions where the operation is between a private acquirer and a listed target (dark blue) and between a listed acquirer and a listed target (light gray). Panel (B) shows the percentage of the total number of transactions where the acquirer is a domestic firm. The vertical line demarcates the pre and post periods. Source: Zephyr Database (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

However, the graphical evidence presented is not totally sufficient since the pattern could respond to factors such as global trends rather than to differential compliance costs. To test whether the regulation analyzed in the previous section drives the patterns observed in Fig. 8, I take a step further and analyze the type of firms that sell their subsidiaries. In particular, I analyze whether firms that sell their subsidiaries are listed firms that operate in industries more exposed to the regulation. I define industry exposure

as the transfers of earnings and royalties made by a particular industry in 2011 (before the regulation took place) normalized by the added value of that industry, used as a proxy to its earnings. The higher the percentage of added value that firms within an industry repatriate, the higher the impact of the ban on transfers on that industry. For instance, before the regulation, the oil and gas industry was transferring abroad 7.83% of its added value, while the textile industry was transferring only 0.31% of its added value.

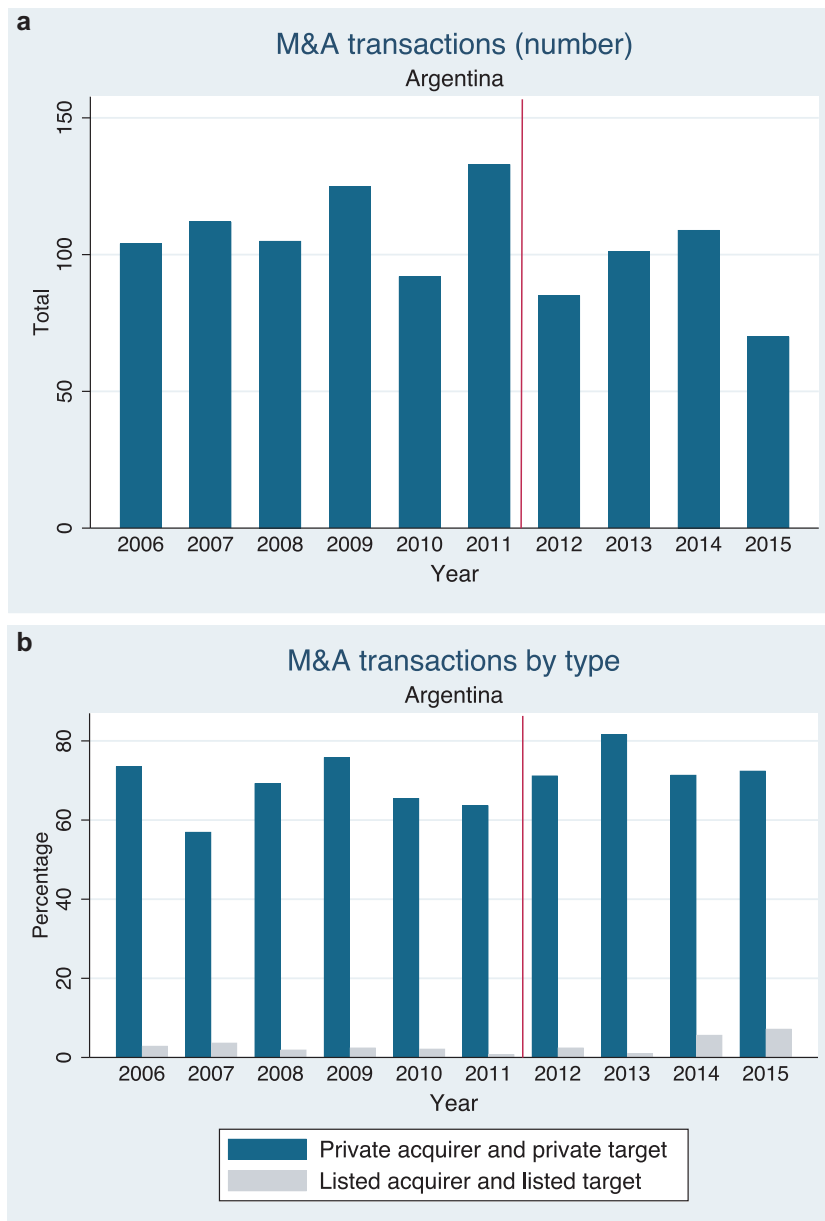


Fig. 9. Transactions before/after ban on international transfers. These graphs plot M&A transactions wherein an MNC sells its Argentine subsidiary, between the years 2006 and 2015. Panel (A) shows the total number of transactions. Panel (B) shows the percentage of M&A Transactions between private acquirers and private targets and between listed acquirers and listed targets. Source: Zephyr Database..

Comparatively, the oil and gas industry was more affected by the regulation than the textile industry. In my analysis, I use the same 18 industry categories that the Central Bank of Argentina uses to report repatriation of earnings and royalties, listed in the Online Appendix, Table A3. In addition, for each industry I report the rank in terms of exposure for 2011 and the two previous years, ranks that suggest the measure of exposure was relatively stable over time. In unreported results, I find that the percentage of M&A transactions that involve firms in highly exposed industries does not vary significantly over time.

For this test, I obtain from Bureau Van Dijk's Orbis database the total number of listed and private MNCs operating within each industry. Then, I calculate the likelihood of selling a subsidiary as the percentage of MNCs operating in a particular industry that sell their subsidiary in year t . To test whether listed firms respond more to the regulation than private firms, I calculate this variable for listed and private MNCs separately. In a difference-in-difference-in-differences (DDD) setting, I exploit three sources of variation: (1) whether the transaction takes place after the regulation, (2) whether the industry is significantly exposed

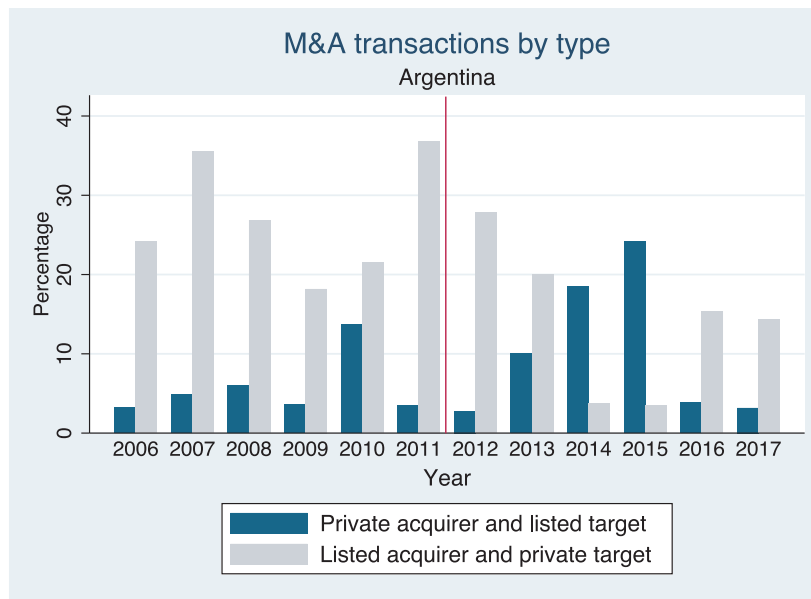


Fig. 10. Type of transactions once ban on international transfers is lifted. This graphs plots M&A transactions wherein an MNC sells its Argentine subsidiary, between the years 2006 and 2017. It shows the percentage of the total number of transactions where the operation is between a private acquirer and a listed target (dark blue) and between a listed acquirer and a listed target (light gray). The vertical lines demarcate the passage and the lifting of the regulation. Source: Zephyr Database (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

to the regulation, and (3) whether the MNC's parent firm is listed. Thus, the empirical specification is as follows:

$$\begin{aligned}
 Sell_{s,l,t} = & \alpha + \beta_h \times HighExposure_s + \beta_p \times Post_t + \beta_l \\
 & \times Listed + \beta_{hp} \times HighExposure_s \times Post_t + \beta_{hl} \\
 & \times HighExposure_s \times Listed + \beta_{pl} \times Post_t \times Listed \\
 & + \beta_{hpl} \times HighExposure_s \times Post_t \times Listed + \epsilon_{s,l,t},
 \end{aligned}
 \quad (2)$$

where s is the sector or industry, l is an indicator set to one for listed firms, and t denotes time or year. $HighExposure_s$ is an indicator that is set to one for industries that are in the top quartile of exposure in 2011 and is set to zero for industries in the bottom quartile of exposure. $Post_t$ is an indicator set to one in years following the ban on profits repatriation, and $Listed$ is an indicator set to one for listed firms. Standard errors are clustered at the industry level to account for possible serial correlation in the error terms. The coefficient of interest is the triple interaction term, which measures the comparative likelihood of selling subsidiaries of listed firms that are highly exposed to the regulation after it was imposed, relative to their private counterparties. The final data set includes 288 industry-MNC type-year observations, which result from the combination of the two types of MNC selling their subsidiaries (listed and private), nine industries (top and bottom quartiles in terms of exposure), over 16 years (2000–2015).

The results of the test are reported in Table 7. In columns [1] and [2], I use the full sample (i.e., all the M&A transactions within an industry regardless of the final stake of the acquirer). For robustness, in columns [3] and [4] I include only those M&A transactions within an industry where the final stake of the acquirer is greater

than or equal to 95%, meaning that the MNC sells the entire Argentine operation. I find that the coefficient for the triple interaction term (β_{hpl}) is positive and significant in all specifications, showing that the regulation has a differential effect on listed than on private firms in the same industries and suggesting that the M&A patterns observed in Argentina after the ban on profits repatriation reflect the differential cost of compliance between listed and private firms. These results are robust to using different thresholds for the final stake of the acquirer.

3.2. The aftermath

At the time of the passage of the regulation, Argentine President Cristina Fernandez de Kirchner had recently been reelected for an additional four-year term. Four years later, in 2015, elections were held to determine her successor, with the two main candidates being Fernandez de Kirchner's ally Daniel Scioli—former vice president under her husband and governor of Buenos Aires for two terms under her—and Mauricio Macri, then mayor of Buenos Aires and the opposition leader. In a contested election, Scioli obtained 37.08% of the votes and Macri 34.15%, leading to a runoff ballotage, in which Macri obtained 51.34% of the votes. In mid-December of 2015—only a week after taking office—Macri eliminated capital controls measures and the ban on profits repatriation.

This episode allows me to further test whether the patterns observed in the M&A in Argentina respond to the differential cost of compliance faced by listed firms. To test this, I expand the database of M&A transactions by including transactions made in the 2016–2017 period. Consistent with the argument of a differential cost of compli-

Table 7

The impact of exposure to a ban on international transfers on firms' selling decision.

	Likelihood of selling			
	(1)	(2)	(3)	(4)
<i>HighExposure_s</i>	0.650** (0.201)		0.291** (0.118)	
<i>Post_t</i>	−0.0121 (0.055)	−0.0121 (0.055)	−0.0669 (0.0444)	−0.0669 (0.0449)
Listed	−0.158 (0.087)	−0.158 (0.088)	−0.104 (0.0642)	−0.104 (0.0650)
<i>HighExposure_s × Post_t</i>	−0.312** (0.111)	−0.312** (0.112)	−0.194** (0.0744)	−0.194** (0.0753)
<i>HighExposure_s × Listed</i>	−0.581** (0.197)	−0.581** (0.200)	−0.263* (0.118)	−0.263* (0.119)
<i>Post_t × Listed</i>	0.0284 (0.063)	0.0284 (0.063)	0.0771 (0.0481)	0.0771 (0.0487)
<i>HighExposure_s × Post_t × Listed</i>	0.256* (0.116)	0.256* (0.117)	0.171** (0.0731)	0.171* (0.0740)
Industry FE	No	Yes	No	Yes
N	288	288	288	288
R-squared	0.468	0.564	0.346	0.436
Sample	Full	Full	Subsample	Subsample

This table presents the regression of the percentage of MNCs in each industry that, after regulation A5264, sold their operations in Argentina based on the exposure of that industry to the regulation. The exposure to the regulation is measured as the percentage of added value that was being transferred abroad before the regulation. The sample period is from 2000 to 2015. Columns [1] and [2] include all M&A transactions where the subsidiary of an MNC is sold. Columns [3] and [4] include the subsample of those transactions where the final stake of the acquirer is equal or higher than 95%. Standard errors are clustered at the industry level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

ance, Fig. 10 shows that the patterns disappear following the lifting of the regulation and the percentage of transactions where operations are between private acquirers and listed targets and between listed acquirers and private targets return to pre-regulation levels.

In addition, I expand the empirical specification in Eq. (2) by including an indicator (*Lifted_t*) that is set to one in the years following the lifting of the ban on profits repatriation and including the corresponding interaction terms. In unreported results, I find that the coefficient on the triple interaction term (*HighExposure_s × Lifted_t × Listed*) is negative—consistent with a reversal in the patterns observed—although the coefficient is small in magnitude (−4.4%) and is marginally insignificant with a *p*-value of 10.4%. This is possibly due to the low power given by the availability of only two years of data following the lifting of the regulation.

4. Conclusion

Over the last two decades, there has been a large decline in the number of firms that decide to go or remain public. In this paper, I uncover a differential cost faced by listed multinational firms that operate in emerging markets and might help explain this pattern, the differential cost of compliance with regulations that target all firms.

I show that, when facing regulations that limit their operations, listed firms comply more than private ones. In addition, I show that private firms can significantly mitigate the regulatory burden presented to them. I then provide

evidence suggesting that the differential compliance cost imposed on listed firms is of such magnitude that it shapes the pattern of M&A transactions in an emerging market. In particular, I demonstrate that when a country becomes more business-unfriendly, private firms acquire listed ones. Meanwhile, listed firms stop acquiring private ones.

The findings in this paper have important implications for managers and policymakers. For the former, this paper uncovers an additional cost faced by listed multinational firms operating in emerging markets: the differential compliance cost associated with regulations that target all firms. Listed firms are subject to stricter monitoring; therefore they cannot bypass regulations nor reduce the burden these regulations impose on them, whereas private firms can and do.

For policymakers, this paper shows that changes in the regulatory environment shape the composition of firms that operate in a country. In particular, business-unfriendly regulations attract firms that defy those regulations, thereby limiting the effectiveness of the policies. Future research might attempt to quantitatively measure this and look at the welfare implications of attracting firms that partially mitigate the effect of regulations.

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